

The School of Mathematics



THE UNIVERSITY
of EDINBURGH

Hierarchical Bayesian Hedonic Pricing of Realised Wholesale Tractor Prices

by

Riccardo Monforte, S2882823

Dissertation Presented for the Degree of
MSc in Statistics with Data Science

August 2026

Contents

Executive Summary	1
1 Background and Objective	2
2 Data	4
2.1 Exploratory analysis	6
3 Methods	11
3.1 INLA	11
3.2 Bayesian hedonic model specification	12
3.2.1 Model 1: fixed-effects hedonic model	13
3.2.2 Model 2: hierarchical random-intercept model	13
3.2.3 Model 3: temporal extension	14
3.2.4 Model 4: joint realised and asking-price model	14
3.2.5 Model 5: robust Student-t specification	15
4 Results	17
4.1 Model comparison and selection	17
4.2 Parameter interpretation and predictive assessment	21
5 Conclusions	26
References	27
Appendices	29
A Data pre-processing	29
A.1 Initial standardisation	29
A.2 Manufacturer names	29
A.3 Model-series identifiers	31
A.4 Transmission categories	31
A.5 GPS categories	32
A.6 Binary equipment variables	32
A.7 Machine age and quantitative validity checks	32
A.8 Construction of the price variables	32
A.9 Construction of the analytical samples	33
A.10 Chronological training and test split	34
AI disclosure	36

List of Tables

1	Explanatory variables, their role within model specification and description.	5
2	Summary statistics for quantitative and binary variables and transaction dates.	8
3	Summary statistics for categorical variables.	8
4	Conditional mean wholesale price and horsepower by categorical and binary variable level. For variables with more than five levels, only the five levels with the highest conditional mean wholesale price are reported.	9
5	Fixed-effect components included in the linear predictor	12
6	Model 1–5 in-sample, out-of-sample and Bayesian performance metrics.	17
7	Posterior summaries of the fixed effects in Model 5.	21
8	Variables retained from the raw dataset	29
9	Examples of manufacturer-name harmonisation	30
10	Final transmission categories	31
11	Final GPS categories	32
12	Validity rules applied to quantitative variables	33
13	Final analytical samples	33
14	Chronological training and test samples	34

List of Figures

1	Distribution of quantitative variables in the sold-price dataset.	6
2	Tractor sale price versus age, operating hours and horsepower.	7
3	Correlation heatmap of quantitative and binary variables.	10
4	Residual-versus-fitted and normal Q-Q plots for Model 3.	18
5	PIT histograms for Models 1-5	19
6	Estimated country, manufacturer and model-series random effects.	22
7	Estimated monthly RW1 effects, showing temporal variation in tractor prices.	23
8	Observed versus fitted training prices and observed versus predicted test prices.	23
9	Observed test prices and corresponding 95% posterior predictive intervals.	24
10	Residual-versus-fitted plot for Model 5.	25

Word Count

4887 words.

Executive Summary

Accurate valuation of used tractors is important for auction houses, dealers and agricultural businesses, but remains challenging because of substantial variation in machine characteristics and market conditions, together with limited visibility of realised transaction prices. Using a large multi-country dataset provided by Spectinga, this dissertation develops a Bayesian hedonic model to predict realised wholesale prices and identify the characteristics most strongly associated with tractor value. Five model specifications were compared, ranging from a fixed-effects model to hierarchical models incorporating manufacturer, model-series, country and temporal effects. An additional model combined realised and advertised asking prices, but this information did not produce uniformly better predictive performance. The largest improvement, instead, resulted from the inclusion of manufacturer, model-series and country effects, which captured important sources of price heterogeneity and substantially improved predictive performance. The results indicate that older and more heavily used tractors are associated with lower prices, whereas greater horsepower is associated with higher values. A key finding is that the price effects of optional equipment vary substantially across features, with front loaders showing a particularly large positive association. The final model explains approximately 92% of in-sample price variation and achieves predictive performance that compares favourably with previous studies of used tractor pricing.

1 Background and Objective

Accurate valuation of used agricultural machinery is especially important for auction houses, dealers, buyers and agricultural businesses in general.

For auction houses, for example, a reliable pricing model can support the definition of guide or reserve prices, identify valuations that appear inconsistent with comparable transactions and provide price ranges for machines with limited sales histories.

Reliable estimates are also relevant to farm managers. Knowing the expected resale value and depreciation of existing machinery can inform decisions about whether to retain or replace an asset and whether the productivity gains, lower maintenance costs or technological improvements offered by a newer tractor justify the investment. Along this line of thought, previous studies have highlighted the importance of machinery values and depreciation in informing replacement and asset-management decisions (Perry and Nixon, 1991; Pearson et al., 2026).

However, valuing used tractors remains a complex task. Tractors are highly differentiated assets traded in relatively opaque markets, where realised transaction prices are often not publicly observable. Consequently, suitable data tend to be fragmented, require extensive validation and cleaning, and are frequently held by private organisations, particularly platforms that intermediate transactions between buyers and sellers. The data used in this study were provided by one such platform, Spectinga, which is gratefully acknowledged for granting access to the dataset.

The valuation of differentiated goods is commonly approached through hedonic pricing models, which explain market prices in terms of the observable characteristics of the goods being traded. Griliches (1961) provided an early empirical application by relating automobile prices to their attributes, while Rosen (1974) developed the formal economic framework in which a differentiated product is represented as a bundle of characteristics and its market price is described by a hedonic price function. Building on this well-established approach, which has also been applied to used agricultural machinery, the present study estimates a hedonic pricing model within a hierarchical Bayesian framework.

Adapting Rosen’s notation to agricultural machinery, the price of tractor (i) sold at time (t) can be expressed as

$$\log(P_{it}) = p(\mathbf{x}_i, \mathbf{m}_{it}),$$

where

$$\mathbf{x}_i = (x_{i1}, \dots, x_{iK})$$

denotes the set of observable characteristics of tractor i, while $\mathbf{m}_{it}$ represents market and transaction specific conditions associated with its sale. An empirical hedonic specification may therefore be written as

$$\log(P_{it}) = \beta_0 + \sum_{k=1}^K \beta_k x_{ik} + \sum_{j=1}^J \gamma_j m_{ijt} + \varepsilon_{it}.$$

where P_{it} is the realised price of tractor i sold at time t , x_{ik} represents its k -th observable characteristic, and m_{ijt} denotes the j -th market or transaction characteristic associated with the sale. In this log-linear specification, the coefficient on a continuous predictor can be interpreted approximately as the percentage difference in price associated with a one-unit increase in that predictor, holding all other variables constant.

Modelling log-price is common in hedonic applications because prices are often right-skewed and the transformation can reduce heteroskedasticity and the influence of very high-priced observations (see, e.g., Griliches, 1961; Agnello and Pierce, 1996; Garay et al., 2022).

Since its early development, hedonic regression has been applied to a remarkably wide range of differentiated goods, including automobiles (Griliches, 1961), residential property (Harrison and Rubinfeld, 1978; Kolbe et al., 2021), artworks (Agnello and Pierce, 1996; Garay et al., 2022), wine and vineyards (Oczkowski, 1994; Cross et al., 2011), personal computers (Pakes, 2003), agricultural machinery (Feuz, 2023; Pearson et al., 2026), and even niche products such as Japanese sake (Saito and Nakatsuma, 2023).

Within the literature on tractor prices, previous studies have generally relied on conventional regression or OLS-based depreciation models and have often focused on specific countries, sales channels or segments of the tractor market. This dissertation builds on this literature by estimating a hedonic model within a hierarchical Bayesian framework using a geographically diverse dataset covering North American and European markets and a broad range of tractor manufacturers, ages, usage levels and technical characteristics. To the best of our knowledge, this is the first application of a hierarchical Bayesian hedonic model to used tractor prices.

A key advantage of hedonic models is their transparent specification and interpretation, which makes it possible to assess the association between each machine or market characteristic and the tractor’s price once the model has been fitted. On the other hand, their main challenge lies in variable selection and the choice of functional form, which are typically guided by previous literature, exploratory analysis, empirical evidence and the information available in the data.

Fettig (1963) provided one of the earliest applications of hedonic pricing to agricultural machinery, identifying horsepower and engine type as major determinants of tractor prices. Subsequent studies identified age and accumulated usage as important predictors of machinery value, often allowing for nonlinear effects (Perry et al., 1990; Perry and Nixon, 1991; Cross and Perry, 1995; Diekmann et al., 2008; Wilson, 2010; Allison Jr et al., 2022; Pearson et al., 2026). Nonlinear relationships between horsepower and price have also been considered (Diekmann et al., 2008; Feuz, 2023). Other relevant characteristics include manufacturer, transmission, drivetrain, cab configuration and hydraulic equipment. Prices may also reflect market and transaction conditions in addition to tractor specific characteristics. These include, for example, the timing and geographical location of the sale and the auction mechanism employed. (Cross and Perry, 1995; Diekmann et al., 2008; Feuz, 2023; Pearson et al., 2026).

The objective of this dissertation is to exploit the Bayesian framework to develop a predictive model for realised wholesale tractor prices, assess whether it can improve on the predictive performance of models previously proposed in the literature, and identify the machine and market characteristics most strongly associated with price variation.

2 Data

The original dataset, `market_listings.csv`, contains 354,607 records across several types of agricultural machinery, of which approximately 79% are tractors. Since this dissertation focuses on tractor pricing, only tractor observations were retained. The raw data include 46 variables describing machine characteristics, sales channel and observation date. Based on previous literature and data availability, 16 substantive variables were selected for analysis and model development. A complete description of the data-preprocessing procedure is provided in Appendix A.

Two analytical datasets were then constructed. The first contains realised wholesale prices and provides the primary outcome used to estimate and evaluate the pricing models. The second contains advertised asking prices and is incorporated in Model 4 to estimate their average relationship with realised wholesale prices, as described in Section 3. Both datasets were divided into training and test subsets: the sold-price dataset was split chronologically into approximately 80% training and 20% test observations. The resulting cutoff date was subsequently applied to the asking-price dataset to ensure that both training samples covered the same historical period.

Age, operating hours, horsepower and equipment characteristics were included as fixed effects, while manufacturer, model series and country were treated as group-level effects. Their hierarchical specification is described in Section 3. Table 1 provides a concise description of the variables used to explain realised wholesale tractor prices.

Table 1: Explanatory variables, their role within model specification and description.

Variable	Role and type	Description
price_sold	Outcome variable	Realised sale price of the tractor. Prices were converted into euros using ECB exchange rates matched to each transaction’s observed_date .
age	Fixed effect; continuous	Number of years since manufacture, obtained as difference between the variable year , indicating the year of tractor production and observed_date , indicating the date in which the tractor was sold.
hours	Fixed effect; continuous	Accumulated engine run-time (the machinery equivalent of a car’s mileage).
horsepower	Fixed effect; continuous	Engine power, main size proxy
gps	Fixed effect; categorical	Level of satellite-guidance equipment, recoded as GPS ready, GPS receiver, Auto steering, RTK or GPS present. These categories distinguish tractors that are only prepared for a guidance system from those equipped with progressively more advanced guidance technology.
transmission	Fixed effect; categorical	Transmission type, encoded as manual/mechanical, CVT/stepless, full powershift, semi-powershift, powershift unspecified, hydrostatic, shuttle/reverser or automatic unspecified.
front_pto	Fixed effect; binary	Presence of a front power take-off used to operate front-mounted implements.
front_loader	Fixed effect; binary	Presence of a front-mounted loader used for lifting and material-handling operations.
front_linkage	Fixed effect; binary	Presence of a front three-point linkage for attaching additional implements.
front_axle_- suspension	Fixed effect; binary	Presence of a suspended front axle, generally associated with improved ride comfort and traction.
cab_suspension	Fixed effect; binary	Presence of a suspended cab, typically found on tractors with higher equipment specifications.
air_conditioning	Fixed effect; binary	Presence of an air-conditioning system in the tractor cab.
isobus	Fixed effect; binary	Compatibility with the ISO 11783 communication standard, which enables communication between tractors and agricultural implements.
make	Group-level effect	Tractor manufacturer, such as John Deere, New Holland or Claas. This effect captures systematic price differences across manufacturers.
model_series	Group-level effect	The manufacturer’s product line, which often reflects differences in size, power class and specification;
country	Group-level effect	The country in which the tractor was sold, capturing geographical differences in prices and market conditions;
source	Model 4 group-level effect	Source from which the asking-price listing was obtained. This variable is used only in Model 4 to capture source-specific differences in average asking-price levels.
Monthly time index	Temporal effect	Monthly index constructed from observed_date and modelled as a structured temporal effect. This is the main “market variable” included in our model. We will verify previous work evidences of seasonality in tractor prices (Pearson et al.2026)

2.1 Exploratory analysis

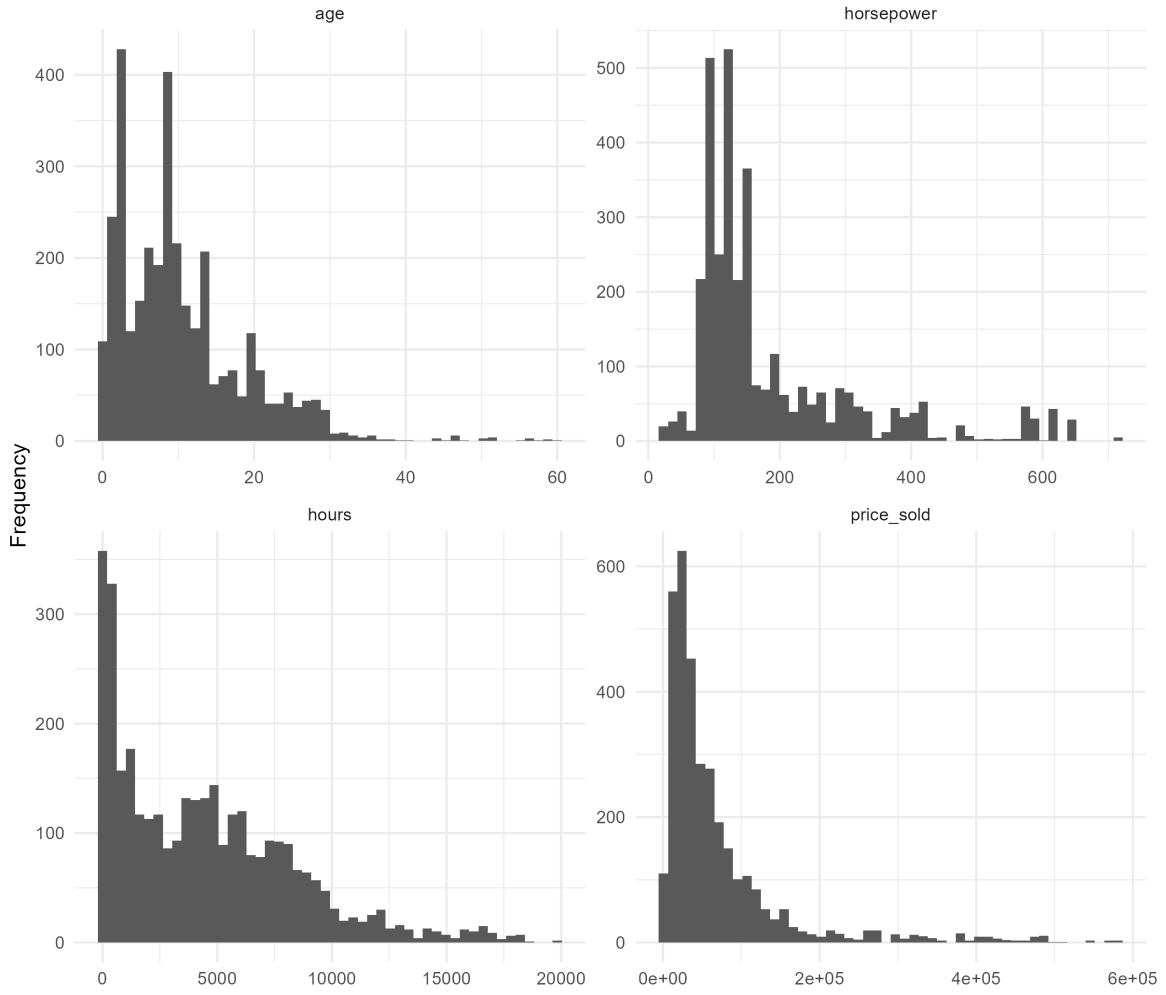


Figure 1: Distribution of quantitative variables in the sold-price dataset.

As a premise, All figures and summary statistics presented in this section are based exclusively on the training subset of the sold-price dataset (test set was excluded to avoid information leakage). Figure 1 presents the distributions of the quantitative variables included in the wholesale-price dataset. Most variables are strongly right-skewed and, in the case of the dependent variable, `price_sold`, the distribution also shows relatively long tails. To reduce the influence of extreme observations, and in line with previous applications of hedonic pricing models, the logarithm of price is used in the analysis.

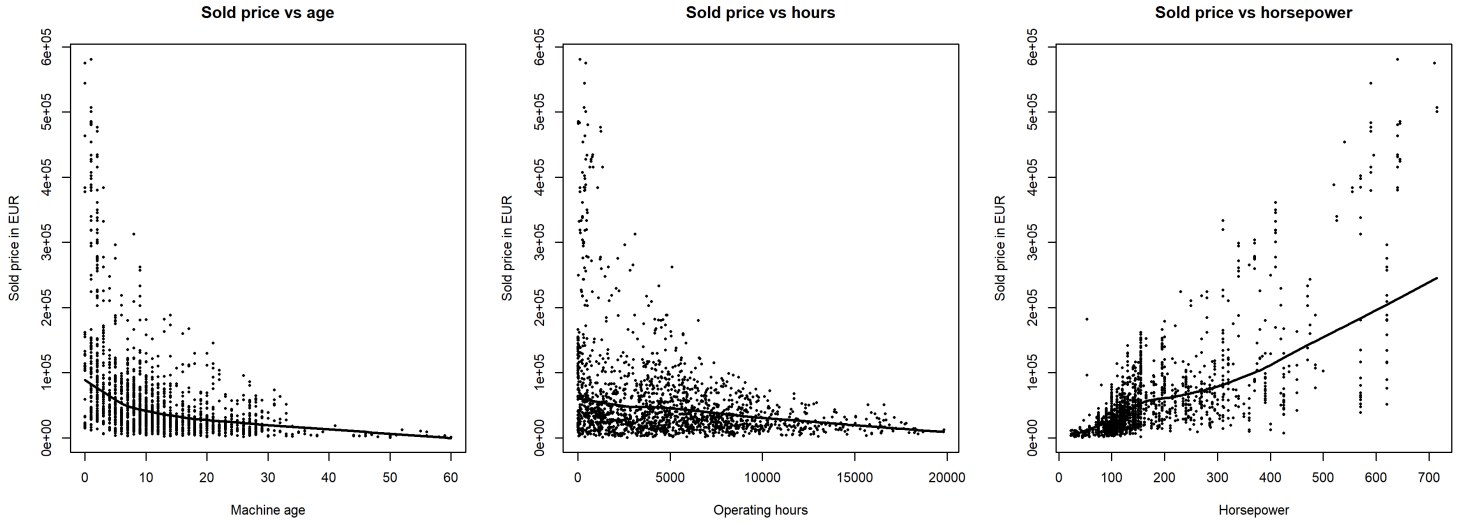


Figure 2: Tractor sale price versus age, operating hours and horsepower.

As a preliminary analysis of the relationship between the target variable and the quantitative predictors, Figure 2 presents three scatterplots relating tractor prices to each quantitative variable considered in the analysis. Price generally decreases with age and operating hours, while its relationship with horsepower is positive and appears nonlinear, becoming steeper at higher power levels. For that reason, both horsepower and its squared term are considered in the subsequent model specification.

Table 2: Summary statistics for quantitative and binary variables and transaction dates.

Variable	Mean	Std. Dev.	Minimum	Maximum
price_sold	68,692.98	84,744.54	507.38	581,239.08
age	10.36	8.49	0	60
hours	4,451.55	3,920.27	1	19,832
horsepower	185.94	131.54	23	715
front_pto	0.07	0.25	0	1
front_loader	0.15	0.35	0	1
front_linkage	0.05	0.21	0	1
front_axle_suspension	0.25	0.43	0	1
cab_suspension	0.26	0.44	0	1
air_conditioning	0.78	0.42	0	1
isobus	0.28	0.45	0	1
observed_date	–	–	2021-12-16	2026-03-28

Table 2 reports summary statistics for the quantitative and binary variables, together with the earliest and latest transaction dates observed in the wholesale-transactions dataset.

Table 3: Summary statistics for categorical variables.

Variable	Levels	Most frequent level	Frequency	Percentage
country	11	Canada	1,231	36.54
make	7	John Deere	1,981	58.80
model_series	465	6155m	169	5.02
transmission	7	Not reported	1,838	54.56
gps	6	Not reported	2,896	85.96

Table 3 reports, for each categorical variable, the number of levels, the most frequent level and its absolute and relative frequency in the wholesale-transactions dataset.

Table 4: Conditional mean wholesale price and horsepower by categorical and binary variable level. For variables with more than five levels, only the five levels with the highest conditional mean wholesale price are reported.

Variable	Level	Mean price (€)	Mean horsepower
make	John Deere	82,708.22	194.64
make	Fendt	76,958.16	214.30
make	Case IH	67,467.13	215.58
make	Claas	40,249.07	138.51
make	Deutz-Fahr	32,746.46	119.79
model_series	9rx710	575,150.94	710.00
model_series	steiger715quadtractrack	504,175.69	715.00
model_series	steiger645quadtractrack	484,070.52	645.00
model_series	9r590	483,805.94	590.00
model_series	9rx590	464,705.59	590.00
country	Canada	81,331.78	196.68
country	United States	74,997.30	195.96
country	Australia	69,062.66	227.26
country	Germany	57,397.39	175.82
country	France	56,522.41	181.52
transmission	Other / unclassified	179,611.65	260.00
transmission	CVT / stepless	93,330.64	225.67
transmission	Full powershift	82,760.52	264.28
transmission	Not reported	63,472.59	168.96
transmission	Powershift unspecified	45,733.94	140.00
gps	RTK	172,889.82	344.57
gps	Auto steering	151,000.52	332.09
gps	GPS present	109,799.28	296.35
gps	GPS receiver	102,910.92	278.44
gps	GPS ready	63,568.63	290.49
front_pto	FALSE	69,080.31	186.43
front_pto	TRUE	63,455.70	179.26
front_loader	FALSE	70,125.72	195.71
front_loader	TRUE	60,354.69	129.06
front_linkage	FALSE	69,675.13	186.01
front_linkage	TRUE	48,189.00	184.56
front_axle_suspension	TRUE	83,881.08	215.67
front_axle_suspension	FALSE	63,680.25	176.13
isobus	TRUE	131,504.20	289.52
isobus	FALSE	44,493.10	146.03
air_conditioning	TRUE	78,508.78	202.21
air_conditioning	FALSE	34,881.56	129.88
cab_suspension	TRUE	113,951.54	237.57
cab_suspension	FALSE	53,131.40	168.19

Table 4 reports the conditional mean wholesale price and horsepower for each level of the categorical variables considered in the analysis. For variables with more than five levels, only the five levels with the highest conditional mean wholesale price are reported. John Deere exhibits the highest mean wholesale price among the manufacturers shown, although this unadjusted value may reflect the composition of tractors observed within each manufacturer rather than an intrinsic brand premium. Nevertheless, it provides a useful descriptive result to consider when interpreting the model estimates reported in Figure 6. For binary variables, TRUE indicates the presence of the characteristic, whereas FALSE indicates its absence.

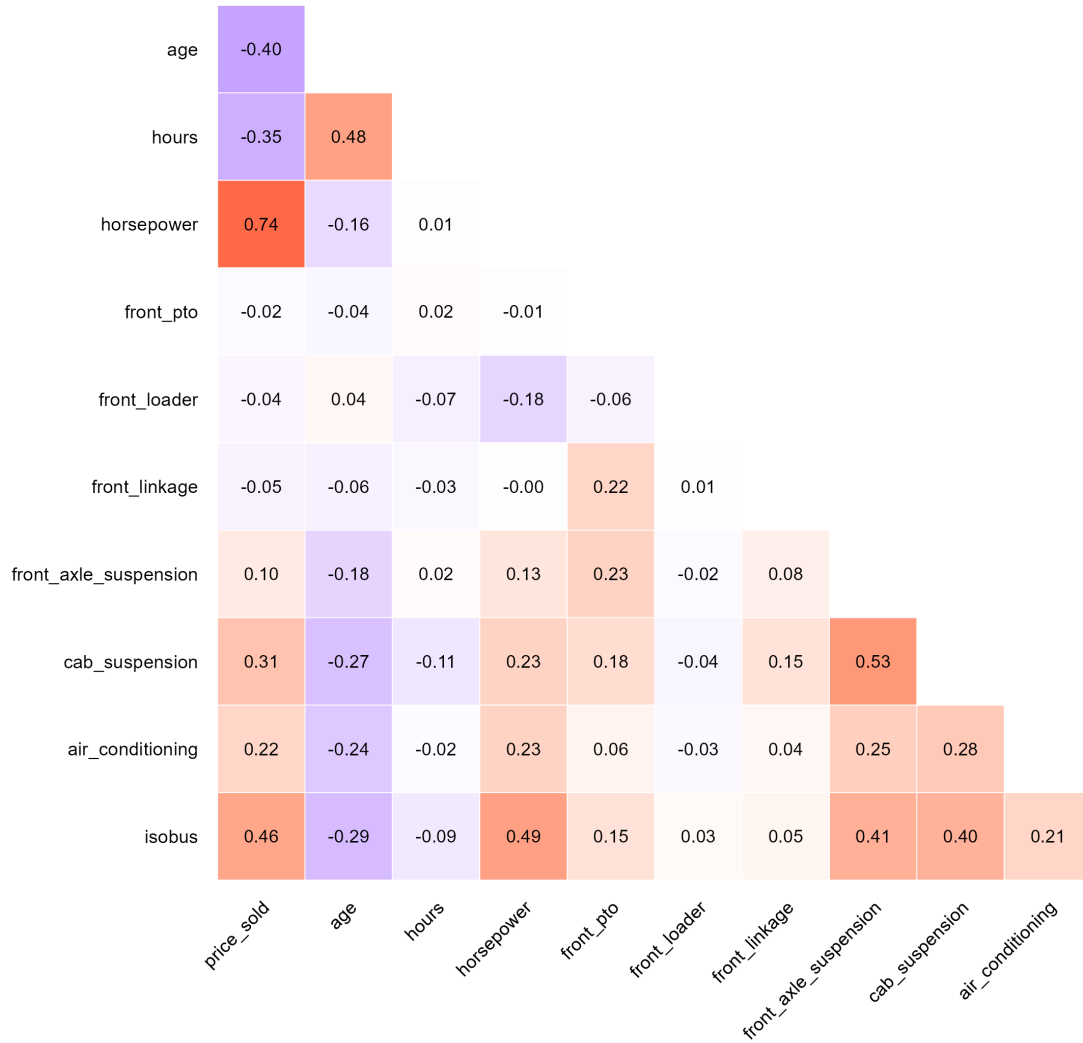


Figure 3: Correlation heatmap of quantitative and binary variables.

Figure 3 reports pairwise correlations among the quantitative and binary variables. Sale price is strongly positively correlated with horsepower ($r=0.74$) and negatively correlated with age ($r=-0.40$) and operating hours ($r=-0.35$), consistent with previous evidence that larger tractors command higher prices, while depreciation increases with age and use. Among the predictors, the largest correlations are observed between front-axle and cab suspension ($r=0.53$), horsepower and ISOBUS ($r=0.49$), and age and operating hours ($r=0.48$). Since these relationships are moderate and the variables represent distinct tractor characteristics, no predictor was excluded a priori.

3 Methods

3.1 INLA

All models were fitted using Integrated Nested Laplace Approximation (INLA), a deterministic approach to Bayesian inference for latent Gaussian models developed by Rue, Martino and Chopin (2009). Conditional on a relatively small set of hyperparameters, the regression coefficients and random effects are jointly assigned a Gaussian prior distribution. INLA exploits the sparse precision structure of the resulting latent field and combines Laplace approximations with numerical integration to efficiently obtain marginal posterior distributions, without requiring the convergence and mixing diagnostics associated with MCMC methods (Rue et al., 2017).

A limitation of INLA is that the numerical integration may become more demanding as the dimension and complexity of the hyperparameter space increase. This became relevant in the robust specification presented in Model 5, where estimating the degrees of freedom of the Student-t likelihood jointly with the other hyperparameters produced numerical instability. The degrees of freedom were therefore fixed by estimating the model over a grid of candidate values and selecting the specification with the lowest DIC. Overall, INLA was particularly suitable for this study because it allowed the hierarchical models to be estimated efficiently. Its conditional Gaussian assumption was also considered appropriate, as there were no strong substantive reasons to adopt non-Gaussian priors for the regression coefficients or random effects, and diffuse Gaussian priors provided a reasonable specification.

3.2 Bayesian hedonic model specification

Let P_i^S denote the realised price of tractor i , and define

$$y_i^S = \log(P_i^S).$$

Let $\mathbf{x}_i$ denote the vector of centred continuous characteristics, quadratic terms, and categorical and binary equipment indicators reported in Table 5. Categorical predictors were represented through dummy variables, with one category omitted as the reference level. The intercept β_0 was specified separately and is therefore not included in $\mathbf{x}_i$.

Table 5: Fixed-effect components included in the linear predictor

Component	Reference category
<i>Continuous and quadratic terms</i>	
age_i^c	Centred continuous variable
hours_i^c	Centred continuous variable
$(\text{hours}_i^c)^2$	Quadratic term
horsepower_i^c	Centred continuous variable
$(\text{horsepower}_i^c)^2$	Quadratic term
<i>Categorical predictors</i>	
transmission	Reference: Manual/mechanical
gps	Reference: Not reported
<i>Binary equipment indicators</i>	
front_pto	Reference: FALSE
front_loader	Reference: FALSE
front_linkage	Reference: FALSE
front_axle_suspension	Reference: FALSE
cab_suspension	Reference: FALSE
air_conditioning	Reference: FALSE
isobus	Reference: FALSE

Notes: Superscript c denotes centring around the corresponding training-sample mean. Quadratic terms were calculated after centring the continuous variables. For categorical and binary predictors, coefficients are interpreted relative to the indicated reference category.

3.2.1 Model 1: fixed-effects hedonic model

The baseline model was specified as

$$y_i^S \mid \eta_i, \tau_\varepsilon \sim \mathcal{N}(\eta_i, \tau_\varepsilon^{-1}),$$

with linear predictor

$$\eta_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta}. \quad (1)$$

The prior distributions were

$$\beta_0 \sim \mathcal{N}(0, 10^{10}), \quad \beta_j \sim \mathcal{N}(0, 10^8), \quad j = 1, \dots, p,$$

and

$$\tau_\varepsilon \sim \text{Gamma}(0.001, 0.001),$$

where the Gamma distribution is parameterised in terms of shape and rate.

Since the continuous covariates were centred, the intercept represents the expected log price for a tractor in the reference categories, at the mean values of the continuous predictors. Diffuse zero-centered Gaussian priors were assigned to the intercept and regression coefficients, with variances 10^{10} and 10^8 , respectively, so that posterior inference was driven primarily by the data. The residual precision was assigned a $\text{Gamma}(0.001, 0.001)$ prior, whose positive support is compatible with a precision parameter and provides a diffuse specification for the residual variability.

3.2.2 Model 2: hierarchical random-intercept model

Model 2 extended Equation (1) by introducing group-specific effects for manufacturer, model series and country:

$$\eta_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{m(i)} + v_{s(i)} + w_{c(i)}. \quad (2)$$

These variables define natural, unordered groups of transactions. Within each grouping variable, the corresponding effects were assumed to be conditionally exchangeable draws from a common Gaussian population distribution:

$$\begin{aligned} u_m \mid \tau_m &\sim \mathcal{N}(0, \tau_m^{-1}), & m = 1, \dots, M, \\ v_s \mid \tau_s &\sim \mathcal{N}(0, \tau_s^{-1}), & s = 1, \dots, S, \\ w_c \mid \tau_c &\sim \mathcal{N}(0, \tau_c^{-1}), & c = 1, \dots, C. \end{aligned}$$

This assumes that manufacturers, model series and countries may differ systematically in their price effects while belonging to a common underlying population. The resulting partial pooling shrinks poorly represented groups more strongly towards the overall mean while preserving differences supported by sufficient data. When the assumed grouping structure is appropriate, partial pooling and shrinkage reduce the uncertainty of group-specific estimates relative to treating each level as an unrelated fixed effect, leading to narrower credible intervals, particularly for groups with few observations. This is especially relevant for manufacturers and model series, whose least represented levels would otherwise be estimated imprecisely. Country was also included as a random effect because transactions

within the same country may share unobserved market, institutional and geographical characteristics, providing a plausible source of group-level dependence. Uncertainty in both the random effects and their hyperparameters is then propagated into posterior inference and prediction. The random effects were constrained to sum to zero for identifiability. Penalised-complexity priors were assigned to their standard deviations:

$$\Pr(\sigma_g > 0.5) = 0.05, \quad \sigma_g = \tau_g^{-1/2}, \quad g \in \{m, s, c\}. \quad (3)$$

Although half-Cauchy priors have been proposed for hierarchical standard deviations (Gelman, 2006), PC priors were preferred as a principled alternative within the INLA framework. They regularise each random component towards the simpler base model $\sigma_g = 0$, while allowing group-level variation when supported by the data (Simpson et al., 2017). Equation (3) assigns 95% prior probability to a group-level standard deviation below 0.5 on the log-price scale, providing regularisation without excluding substantial price heterogeneity across groups.

3.2.3 Model 3: temporal extension

Model 3 extended Equation (2) by introducing a latent monthly effect:

$$\eta_i = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{m(i)} + v_{s(i)} + w_{c(i)} + f_{t(i)}.$$

Previous research indicates that used agricultural machinery prices are influenced not only by machine characteristics but also by market timing and macroeconomic conditions (Diekmann et al. 2008, Perry et al. 1990; Perry and Nixon 1991, Pearson et al., 2026). The temporal component was therefore included to capture unobserved market conditions shared by transactions observed in the same period, such as changes in aggregate demand, inflation, financing conditions or sector-specific shocks.

The temporal effect was modelled as a first-order random walk:

$$f_t - f_{t-1} \mid \tau_t \sim \mathcal{N}(0, \tau_t^{-1}), \quad t = 2, \dots, T.$$

This specification assumes that market conditions evolve gradually, with effects in adjacent months expected to be more similar than those observed further apart in time. The temporal component was scaled and constrained to sum to zero for identifiability.

A penalised-complexity prior was assigned to its standard deviation:

$$\Pr(\sigma_t > 0.20) = 0.05, \quad \sigma_t = \tau_t^{-1/2}.$$

This specification assigns 95% prior probability to a standard deviation of the scaled temporal effect below 0.20 on the log-price scale, without imposing a strict upper bound. An individual temporal effect of 0.20 corresponds approximately to a 22% increase in price on the original scale, while an effect of -0.20 corresponds to an 18% decrease. The prior therefore favours moderate temporal variation while allowing larger market-driven changes when supported by the data.

3.2.4 Model 4: joint realised and asking-price model

Model 4 extended the previous specification by jointly modelling realised and asking prices. The purpose was to use the larger asking-price sample to inform the common hedonic structure, while allowing the two price types to differ in their average level and residual variability.

For $r \in \{S, A\}$, where S denotes realised prices and A asking prices, let

$$y_i^r = \log(P_i^r).$$

Two separate Gaussian likelihoods were specified:

$$y_i^r \mid \eta_i^r, \tau_{\varepsilon,r} \sim \mathcal{N}(\eta_i^r, \tau_{\varepsilon,r}^{-1}), \quad r \in \{S, A\},$$

with linear predictor

$$\eta_i^r = \beta_0 + \mathbf{x}_i^\top \boldsymbol{\beta} + u_{m(i)} + v_{s(i)} + w_{c(i)} + f_{t(i)} + I(r = A) (\alpha_A + q_{d(i)}).$$

The fixed effects and the manufacturer, model-series, country and temporal components were shared across the two likelihoods. This parsimonious specification assumes that machine characteristics and the common market structure affect realised and asking prices in the same way, allowing the larger asking-price sample to inform the shared hedonic relationship.

The coefficient α_A captures the adjusted average difference between asking and realised log-prices, conditional on the remaining model components. As a fixed effect, it was assigned the same diffuse Gaussian prior as the other regression coefficients:

$$\alpha_A \sim \mathcal{N}(0, 10^8).$$

Since the two datasets contain different listings and each listing records only one price type, α_A should be interpreted as the adjusted average difference between asking and realised prices, rather than as the markup applied to the same tractor.

The term $q_{d(i)}$ captures additional heterogeneity across the sources of asking-price listings:

$$q_d \mid \tau_q \sim \mathcal{N}(0, \tau_q^{-1}), \quad d = 1, \dots, D.$$

This source-specific effect was included only for asking-price observations, allowing sources to differ in their average asking-price levels because of differences in pricing practices or market composition. The effects were constrained to sum to zero and assigned the PC prior

$$\Pr(\sigma_q > 0.20) = 0.05, \quad \sigma_q = \tau_q^{-1/2}.$$

Finally, separate residual precisions were estimated:

$$\tau_{\varepsilon,S}, \tau_{\varepsilon,A} \sim \text{Gamma}(0.001, 0.001),$$

allowing unexplained variability to differ between realised and asking prices. All remaining priors were unchanged from Models 1–3.

3.2.5 Model 5: robust Student-t specification

Model 5 retained the linear predictor and prior specification of Model 3 but replaced the Gaussian likelihood with a Student-t distribution:

$$y_i^S \mid \eta_i, \tau_\varepsilon, \nu \sim t_\nu(\eta_i, s_\varepsilon),$$

where

$$s_{\varepsilon}^2 = \frac{\nu - 2}{\nu \tau_{\varepsilon}},$$

so that the conditional residual variance remains τ_{ε}^{-1} . All fixed, hierarchical and temporal components were unchanged from Model 3.

Compared with the Gaussian likelihood, the Student-t distribution has heavier tails and therefore assigns greater probability to large residuals. Consequently, observations with unusually high or low prices exert less influence on parameter estimation, making the model more robust to extreme observations.

Joint estimation of the degrees of freedom ν with the remaining hyperparameters produced numerical instability. The model was therefore fitted over a grid of fixed candidate values (for ν ranging from 3 to 15), and $\nu=7$ was selected because it yielded the numerically stable model with lowest DIC. This value retains heavier tails than the Gaussian distribution while preserving a finite residual variance. The prior for τ_{ε} and all other priors were unchanged from Model 3.

4 Results

4.1 Model comparison and selection

Table 6: Model 1–5 in-sample, out-of-sample and Bayesian performance metrics.

Model	Train R^2	Test R^2	Train RMSE	Test RMSE	Train MAE	Test MAE	Test MAPE	Test MedAPE	DIC	WAIC	NLSCPO	Marginal log-likelihood
Model 1	0.818	0.290	0.431	0.435	0.329	0.335	35.090	27.340	3944.700	3945.700	1972.900	-2308.400
Model 2	0.925	0.554	0.276	0.345	0.193	0.252	24.760	17.360	1686.100	1677.400	874.100	-1442.000
Model 3	0.930	0.563	0.268	0.342	0.187	0.253	25.740	18.580	1536.600	1534.100	806.100	-1459.300
Model 4	0.902	0.574	0.316	0.337	0.233	0.257	25.850	19.830	–	–	–	–
Model 5	0.926	0.549	0.275	0.347	0.178	0.252	26.350	17.910	917.800	954.500	477.800	-1253.300

Table 6 compares the five model specifications using measures of in-sample fit, out-of-sample predictive accuracy and Bayesian model adequacy. Higher R^2 values indicate that a larger proportion of the variability in log prices is explained, whereas lower RMSE, MAE, MAPE and MedAPE values indicate smaller prediction errors. MAPE and MedAPE are reported on the original EUR price scale; MedAPE was used because it is less sensitive to outliers than MAPE. DIC and WAIC measure the trade-off between model fit and complexity, while NLSCPO evaluates leave-one-out predictive performance. For all three criteria, lower values are preferred. Conversely, a larger, or less negative, marginal log-likelihood indicates greater overall evidence in favour of the model.

It can be noticed that test-set R^2 values are considerably lower than their training counterparts and this may initially suggest limited generalisation. However, this decline is not accompanied by a comparably large deterioration in RMSE and MAE, which remain of a similar order of magnitude across the two samples. This indicates that the lower test R^2 is not solely driven by poorer absolute predictive accuracy. Rather, the chronological test set displays lower variability in log prices, meaning that comparable prediction errors account for a larger proportion of the total variation. Test R^2 should therefore be interpreted jointly with RMSE, MAE, MAPE and MedAPE, which provide complementary evidence that the hierarchical models retain meaningful out-of-sample predictive performance.

The progression from Model 1 to Model 2 provides the clearest evidence of the importance of the hierarchical structure. Model 1, which includes only fixed effects, explains 81.8% of the variation in the training data but only 29.0% in the chronological test set. After introducing manufacturer, model-series and country specific random intercepts, the training R^2 increases to 0.925 and the test R^2 to 0.554. The test RMSE falls from 0.435 to 0.345, while the test MAPE decreases from 35.09% to 24.76%. The improvement is even more pronounced in the Bayesian criteria: DIC falls from 3944.7 to 1686.1, WAIC from 3945.7 to 1677.4, and NLSCPO from 1972.9 to 874.1. These results show that substantial price variation remains after controlling for the observed tractor characteristics and that explicitly modelling the grouping structure considerably improves both fit and generalisation.

The estimated intraclass correlations from the final hierarchical specification reinforce this conclusion. Excluding the temporal component, model series accounts for approximately 44.8% of the total variance, country for 15.1% and manufacturer for 12.6%. Together, these components represent approximately 72.5% of the variance included in the ICC calculation. The particularly large model-series contribution suggests that tractors belonging to the same product family share important price characteristics that are not fully captured by the observed technical predictors. The country and manufacturer effects also remain substantial, supporting their inclusion as separate hierarchical components.

Model 3 extends this structure by adding a monthly first-order random walk. This produces a further, although smaller, improvement, in line with the results of previous studies (eg, see Pearson et al. 2026). The test R^2 rises from 0.554 to 0.563, while DIC, WAIC and NLSCPO fall to 1536.6, 1534.1 and 806.1, respectively. The comparatively modest change in point-prediction metrics indicates that the hierarchical intercepts explain most of the improvement over Model 1.

Model 4 incorporates asking prices through a second likelihood. It achieves the highest test R^2 (0.574)

and the lowest test RMSE (0.337). Relative to the other specifications, it includes an additional fixed coefficient, α , capturing the average difference between asking and realised wholesale prices. Its posterior mean is 0.482, with a 95% credible interval of 0.441–0.524. On the original price scale, this corresponds to asking prices being approximately 61.9% higher than realised prices for otherwise comparable tractors, with an estimated range of 55.4%–68.9%. This suggests a substantial and systematic asking-price premium over wholesale transaction prices.

However, the improvement in predictive performance relative to Model 3 is modest, and the percentage-error measures are not uniformly better. Model 4 was therefore not selected as the final specification. DIC, WAIC, NLSCPO and the marginal log-likelihood are not reported in table 6 because the model is estimated using a different response structure and a larger joint dataset, making these criteria not directly comparable with those of the realised-price-only models.

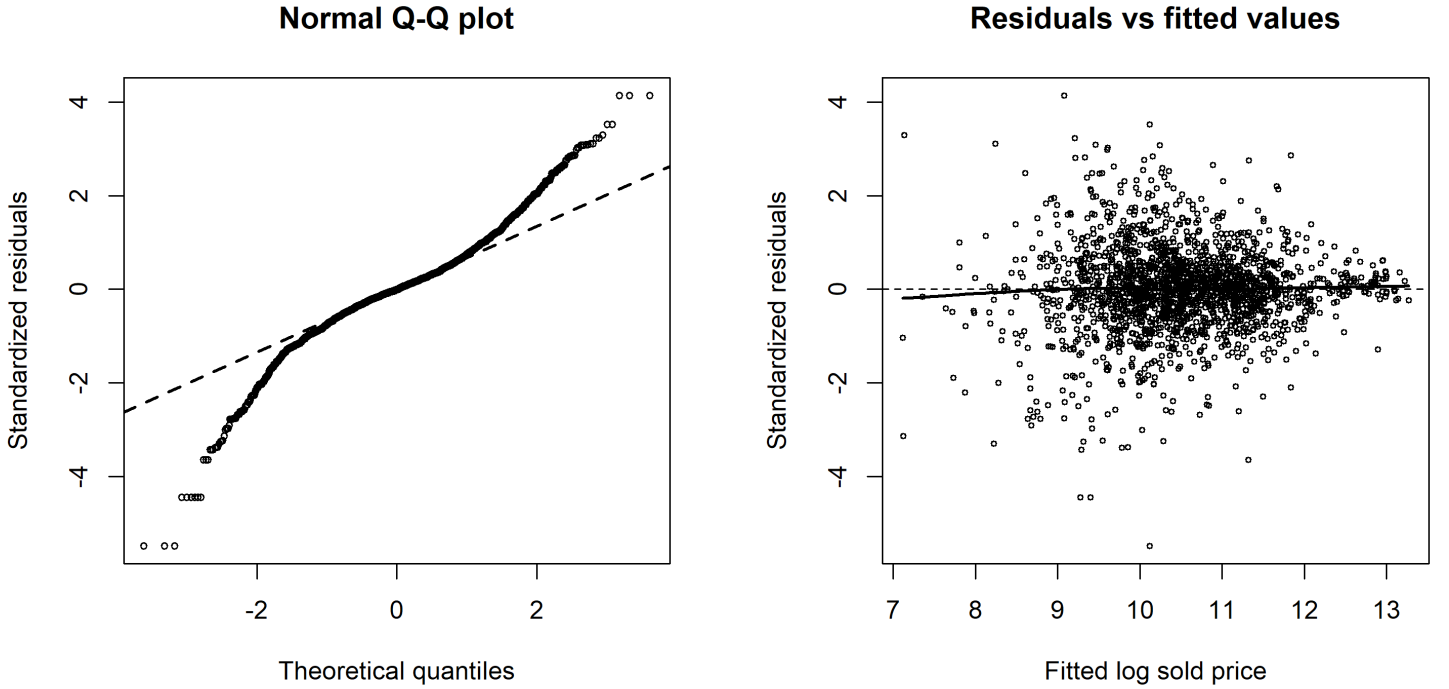


Figure 4: Residual-versus-fitted and normal Q-Q plots for Model 3.

Inspection of the Model 3 residuals (figure 4) motivated the final change in specification that led to Model 5. The residual-versus-fitted plot shows that the smoothed residual trend remains close to zero, suggesting that the conditional mean structure is broadly adequate. However, the normal Q–Q plot displays pronounced deviations in both tails and several extreme observations. The main remaining inadequacy therefore appears to concern the Gaussian likelihood rather than the systematic component of the model.

Model 5 retains the same fixed effects, random effects and temporal structure as Model 3, but replaces the Gaussian likelihood with a Student-t distribution with seven degrees of freedom, allowing extreme observations to receive less influence compared with the Gaussian likelihood. Relative to Model 3, DIC decreases from 1536.6 to 917.8, WAIC from 1534.1 to 954.5 and NLSCPO from 806.1 to 477.8. The marginal log-likelihood also improves from -1459.3 to -1253.3. At the same time, out-of-sample point accuracy remains broadly comparable: Model 5 obtains a test R^2 of 0.549, a test RMSE of 0.347, a test MAE of 0.252 and a MedAPE of 17.91%. Thus, the substantial improvement in probabilistic fit is achieved with only a small reduction in point-prediction performance.

Leave-one-out PIT distributions for realised prices
The dashed line represents the Uniform(0,1) density

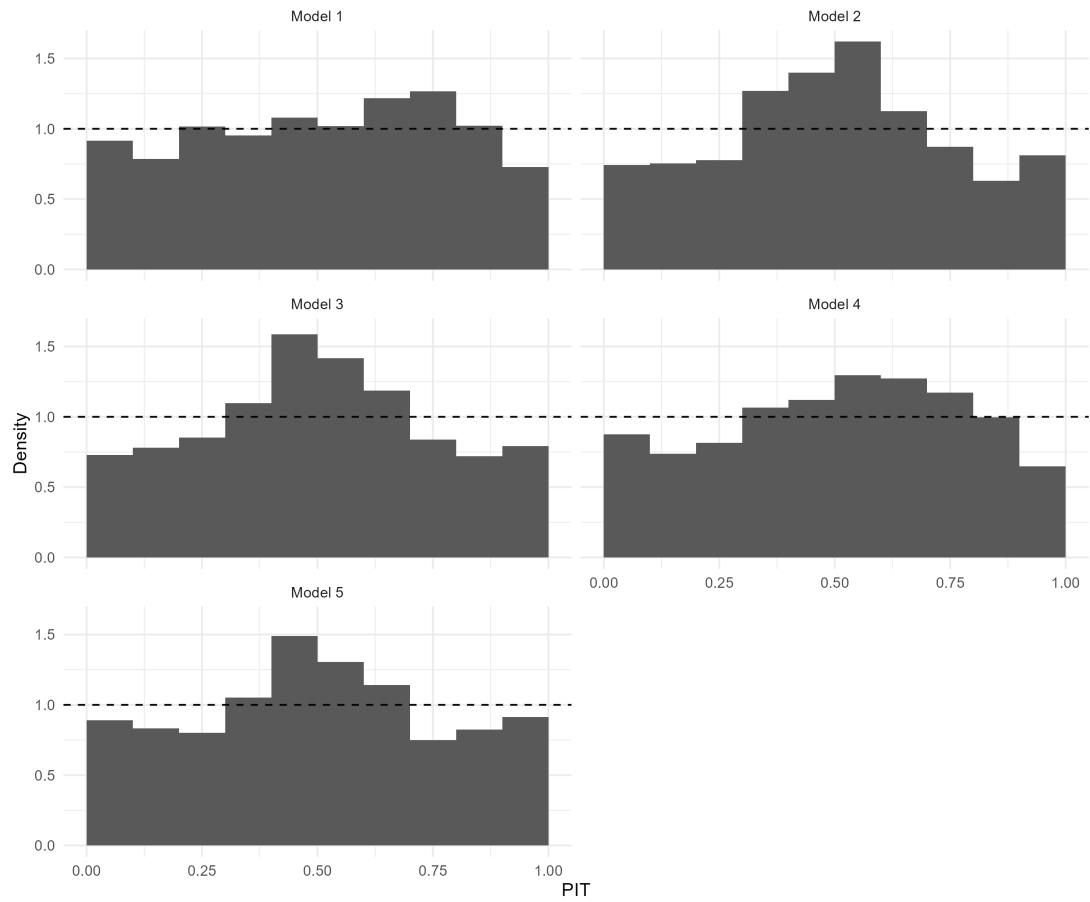


Figure 5: PIT histograms for Models 1-5

The PIT histograms show no pronounced asymmetry, suggesting limited systematic over- or under-prediction. The hierarchical models nevertheless display some concentration around the centre of the unit interval, indicating mildly overdispersed predictive distributions rather than perfect calibration. This pattern remains visible in Model 5, so the robust likelihood improves the treatment of extreme observations without fully eliminating predictive overdispersion.

Overall, Model 5 is selected as the final specification. It preserves the parsimonious linear predictor of Model 3, directly addresses the heavy-tailed residual behaviour and provides by far the strongest DIC, WAIC, NLSCPO and marginal log-likelihood among the directly comparable models, while maintaining competitive test-set accuracy. Models 2, 3 and 5 all obtain training R^2 values above 0.92; to the best of our knowledge, these exceed the in-sample explanatory performance reported in the used-tractor hedonic pricing studies reviewed in this dissertation.

4.2 Parameter interpretation and predictive assessment

Table 7: Posterior summaries of the fixed effects in Model 5.

Parameter	Mean	Std. Dev.	2.5%	Median	97.5%
Intercept	10.454	0.095	10.267	10.454	10.640
Age	-0.0379	0.0016	-0.0410	-0.0379	-0.0348
Hours	-8.37×10^{-5}	2.56×10^{-6}	-8.88×10^{-5}	-8.37×10^{-5}	-7.87×10^{-5}
Hours ²	2.80×10^{-9}	3.01×10^{-10}	2.21×10^{-9}	2.80×10^{-9}	3.39×10^{-9}
Horsepower	0.00762	0.00022	0.00718	0.00762	0.00806
Horsepower ²	-1.08×10^{-5}	7.44×10^{-7}	-1.22×10^{-5}	-1.08×10^{-5}	-9.31×10^{-6}
Transmission: CVT / stepless	0.151	0.090	-0.025	0.151	0.326
Transmission: Full powershift	0.084	0.089	-0.090	0.084	0.258
Transmission: Semi-powershift	0.117	0.089	-0.058	0.117	0.292
Transmission: Powershift unspecified	0.154	0.253	-0.341	0.154	0.650
Transmission: Other / unclassified	0.247	0.147	-0.041	0.247	0.535
Transmission: Not reported	0.087	0.087	-0.084	0.087	0.258
GPS present	0.054	0.040	-0.024	0.054	0.133
GPS ready	-0.050	0.048	-0.144	-0.050	0.045
GPS receiver	0.013	0.037	-0.059	0.013	0.086
Auto steering	0.091	0.021	0.049	0.091	0.132
RTK	0.015	0.046	-0.075	0.015	0.104
Front PTO	0.048	0.022	0.006	0.048	0.090
Front loader	0.335	0.016	0.304	0.335	0.365
Front linkage	-0.031	0.030	-0.089	-0.031	0.027
Front axle suspension	-0.004	0.017	-0.036	-0.004	0.029
Cab suspension	-0.008	0.018	-0.042	-0.008	0.027
Air conditioning	0.092	0.017	0.059	0.092	0.125
ISOBUS	-0.019	0.015	-0.049	-0.019	0.011

Table 7 reports the posterior summaries of the fixed effects in Model 5. Since the continuous covariates were centred, the intercept corresponds to conditional median price of approximately €34,700 for a tractor with average age, usage and horsepower, belonging to the reference categories and with group-level and temporal effects equal to zero. Each additional year of age is associated with a 3.7% reduction in price, with a 95% credible interval of 3.4%–4.0%.

The effects of usage and horsepower are nonlinear. Relative to a tractor at the mean number of operating hours, an additional 1,000 hours is associated with an estimated price reduction of approximately 7.8%. The positive quadratic term indicates that this depreciation becomes progressively less pronounced at higher usage levels. This is consistent with previous evidence that the marginal reduction in value associated with an additional hour declines as tractors accumulate usage (Perry et al., 1990; Perry and Nixon, 1991; Pearson et al., 2026). Conversely, an increase of 100 horsepower from the sample mean is associated with an estimated price increase of approximately 92.4%. The negative quadratic term indicates diminishing price returns to horsepower, consistent with previous evidence that the relationship between tractor size and value is not strictly linear.

Among the equipment characteristics, the largest association is observed for front loaders, which are associated with prices approximately 39.8% higher than those of otherwise comparable tractors, with a 95% credible interval of 35.6%–44.1%. Automatic steering, air conditioning and front PTO are associated with premiums of approximately 9.5%, 9.6% and 4.9%, respectively, and their credible

intervals exclude zero. By contrast, the intervals for the remaining GPS, transmission and equipment indicators include zero, providing limited evidence of systematic price differences after controlling for the other model components.

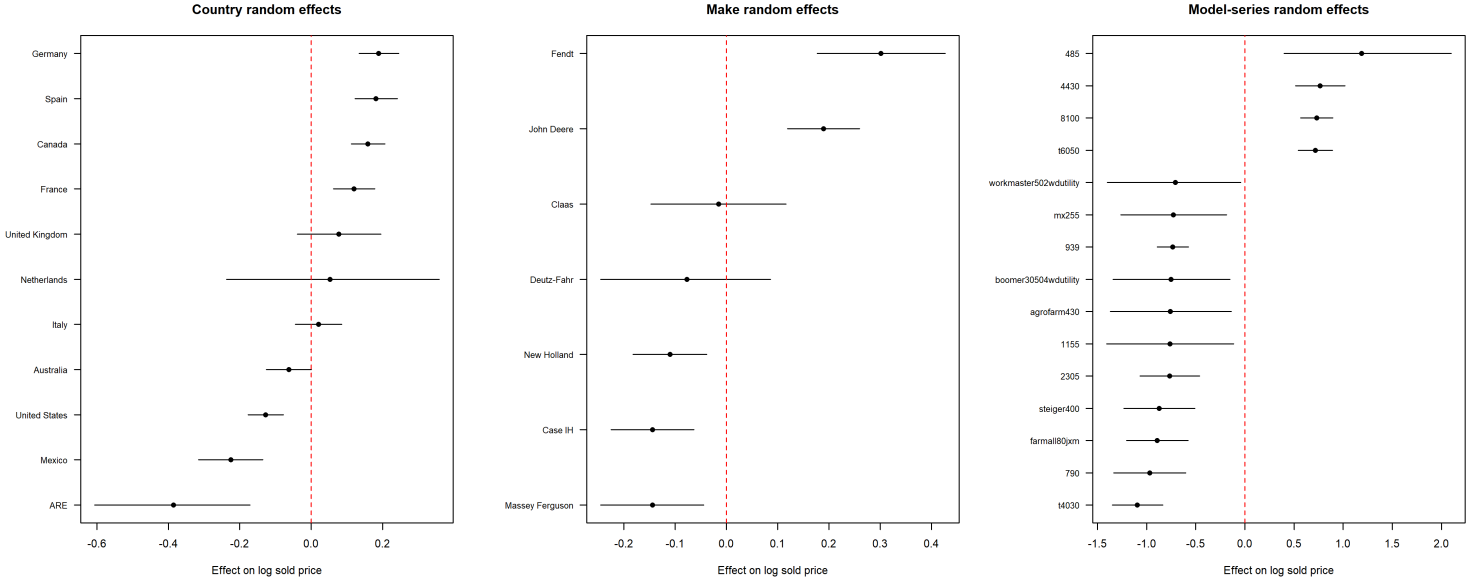


Figure 6: Estimated country, manufacturer and model-series random effects.

Figure 6 displays the estimated random effects for country, manufacturer and model series. Where more than 15 levels were available, the figure reports the 15 levels with the largest absolute posterior means. Each random effect represents a deviation from the model-implied conditional price level associated with a group effect of zero, holding all observed characteristics and the remaining model components constant. Positive values therefore indicate a conditional price premium, whereas negative values indicate a conditional discount.

For country effects, Germany, Spain, Canada and France are associated with systematically higher realised prices. By contrast, the United States, Mexico and the United Arab Emirates are associated with negative effects compared to the baseline level of no group effect. The credible intervals for the other displayed countries overlap zero, providing limited evidence of systematic price differences.

For manufacturers, Fendt and John Deere exhibit positive price effects, while New Holland, Case IH and Massey Ferguson show negative deviations from the overall level. A similar pattern is reported by Pearson et al. (2026), who find the largest positive brand effect for John Deere and the largest negative effect for New Holland. The credible intervals for Claas and Deutz-Fahr overlap zero, indicating insufficient evidence of a systematic price premium or discount.

Model-series effects are considerably larger, ranging from approximately -1.1 to 1.3 log points among the displayed levels. On the original price scale, this corresponds to conditional expected prices ranging from approximately 66.7% below to 266.9% above the zero-effect model-series baseline. This pronounced heterogeneity is consistent with the model-series ICC of 44.8%, suggesting that product families capture substantial unobserved variation in design, market positioning and original value.

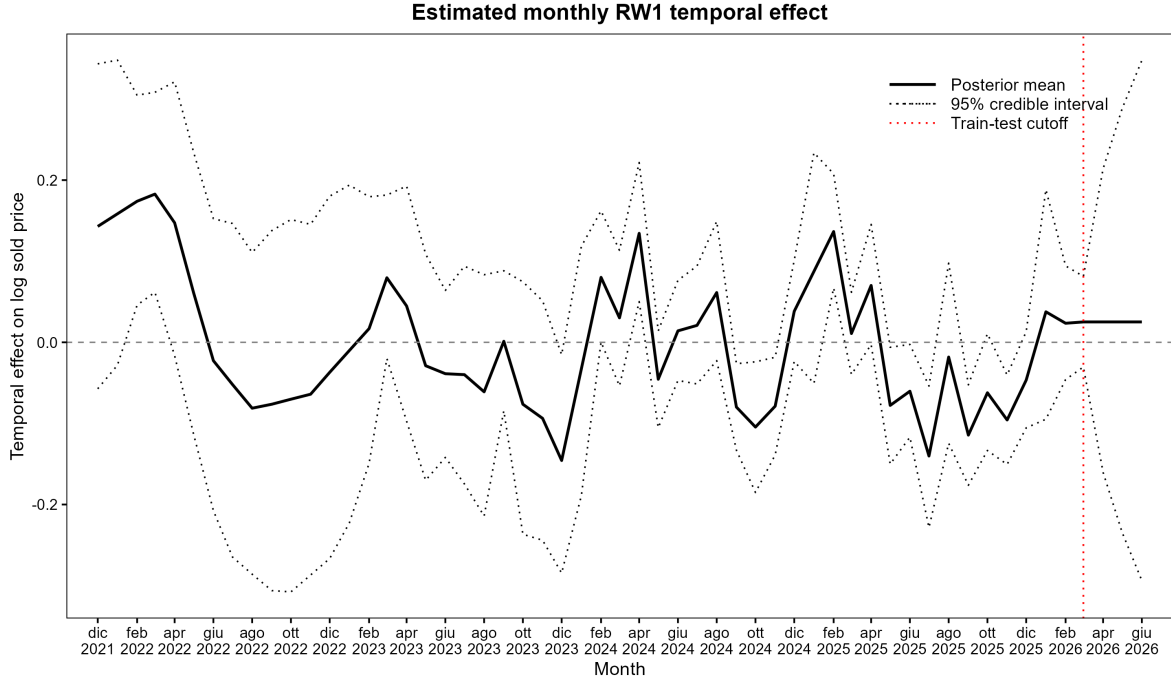


Figure 7: Estimated monthly RW1 effects, showing temporal variation in tractor prices.

The monthly RW1 effect provides further evidence of temporal variation in tractor prices. Posterior mean effects are generally lowest between September and November, ranging from approximately -0.10 to -0.15, which corresponds to prices about 10%–14% below the overall level. Conversely, prices tend to peak between January and March, when the estimated effects approach 0.15–0.20, equivalent to premiums of roughly 16%–22%. Pearson et al. (2026) similarly find that tractor prices are highest during the first three months of the year and lowest during periods of intensive field activity, particularly planting and the September–October harvest. Previous studies have linked this seasonality to both the agricultural production calendar and features of US tax policy (Hadrich et al., 2013; Reid and Bradford, 1983; Allison Jr et al., 2022). However, the emergence of a comparable pattern in the present multi-country sample (approximately 60% of which consists of non-US transactions) suggests that agricultural production cycles are likely to be the dominant mechanism.

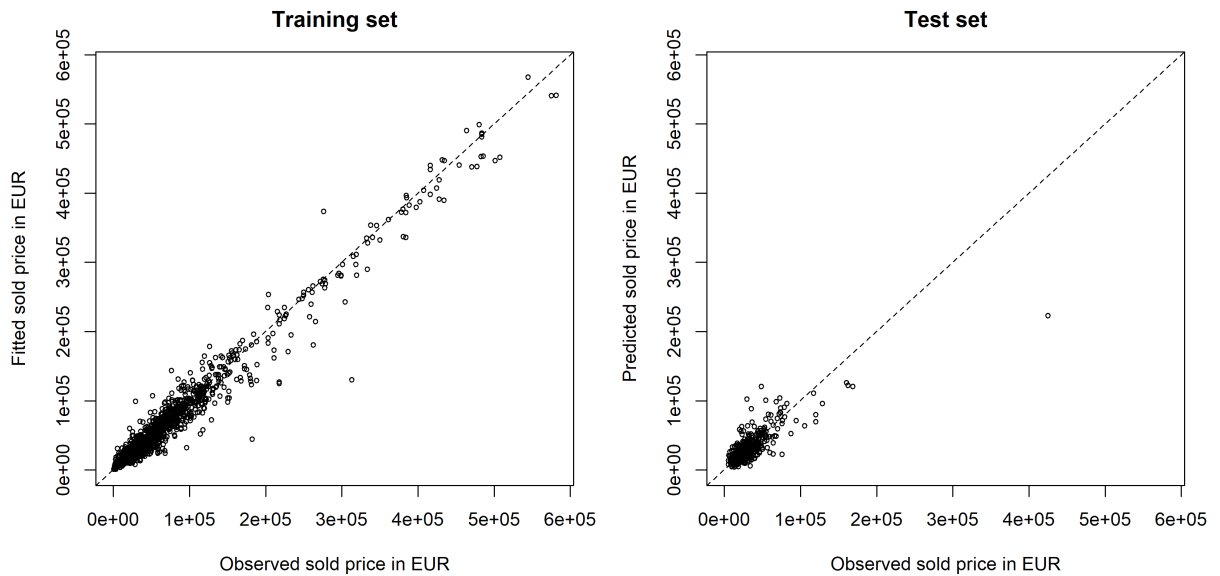


Figure 8: Observed versus fitted training prices and observed versus predicted test prices.

Figure 8 presents the observed-versus-fitted plot for the training set on the left and the observed-

versus-predicted plot for the test set on the right. In both panels, the 45-degree line represents perfect agreement between observed and estimated prices. The fitted values in the training set align closely with the observed prices. Out-of-sample predictions also capture the overall relationship reasonably well, although one high-value observation lies substantially away from the reference line, possibly representing an isolated tractor with exceptional characteristics not fully captured by the model.

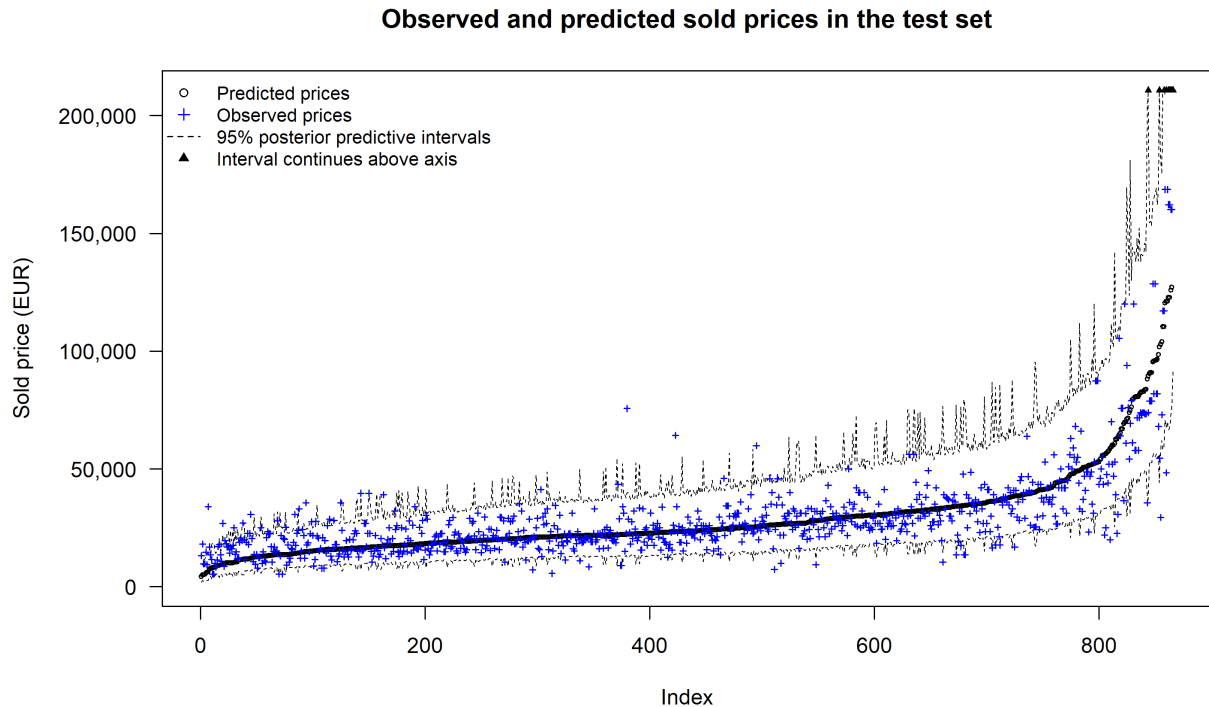


Figure 9: Observed test prices and corresponding 95% posterior predictive intervals.

Figure 9 presents the 95% posterior predictive intervals for test-set tractor prices on the original price scale. The model generally tracks realised prices well, with most observed values falling within their corresponding predictive intervals. The intervals widen for higher-priced tractors, reflecting the greater predictive uncertainty associated with these observations. For visual clarity, the upper ends of a small number of exceptionally wide intervals are truncated at the displayed axis limit and indicated by triangles.

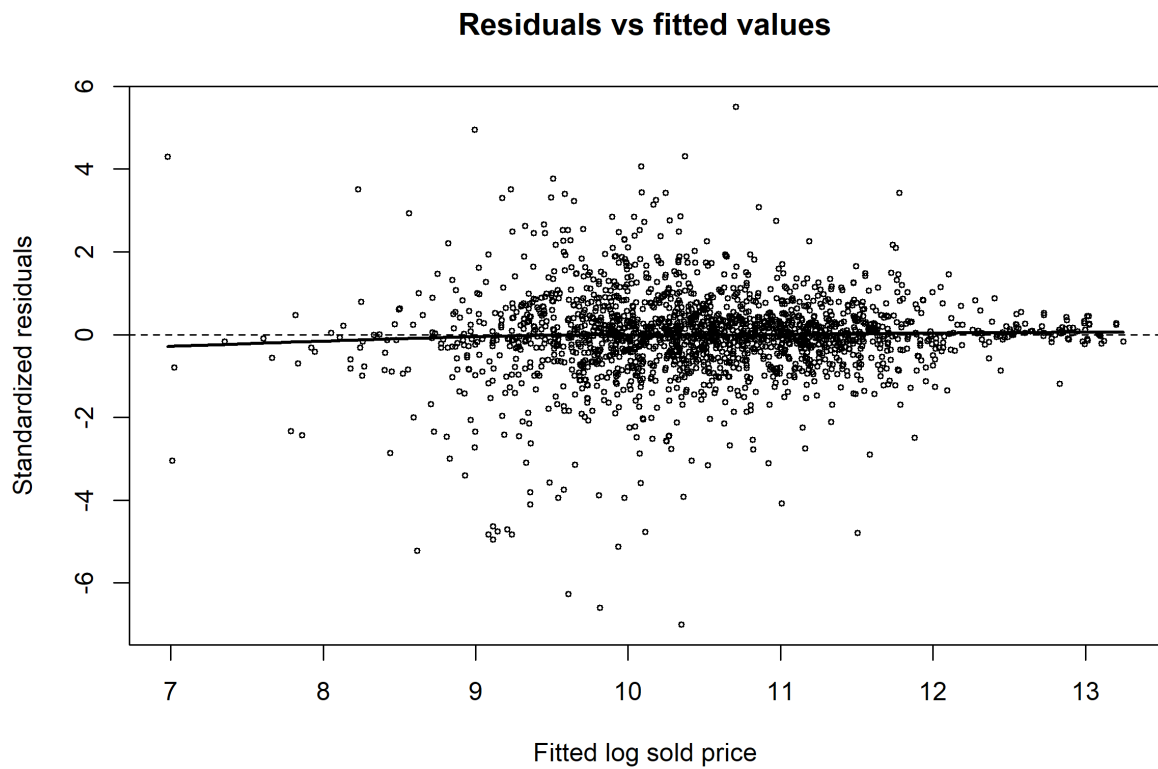


Figure 10: Residual-versus-fitted plot for Model 5.

The residual-versus-fitted (figure 10) plot shows no clear systematic pattern, and the smoothed line remains close to zero across the fitted-value range. This provides further evidence that the conditional mean structure of Model 5 is robust.

5 Conclusions

This dissertation developed a hierarchical Bayesian hedonic model to predict realised wholesale tractor prices and identify the machine and market characteristics most strongly associated with their variation. Five specifications were compared. The largest improvement in out-of-sample performance followed the introduction of manufacturer, model-series and country effects, while the Student-t specification provided the strongest probabilistic fit and reduced the influence of extreme observations. Prices were lower for older and more heavily used tractors and higher for more powerful machines. The effects of optional equipment varied substantially, with front loaders showing the largest positive association.

A first limitation concerns data quality. The original data required extensive cleaning and manual harmonisation, particularly for manufacturer, model-series, transmission and GPS information. Although these procedures made the variables suitable for analysis, some residual classification errors may remain.

A second limitation relates to the asking-price analysis. Asking and realised prices could not be matched for the same tractor, meaning that the estimated 62% markup represents an average relationship between the two markets rather than the outcome of individual negotiations. The corresponding model was also based on a different response structure and could not be compared directly with the remaining specifications using all Bayesian criteria.

Future research could allow the value of tractor characteristics to vary over time, incorporate additional market and macroeconomic variables, and combine matched listing-to-sale records with buyer-engagement measures such as views, enquiries and bids.

References

- Agnello, R. J. & Pierce, R. K. (1996), 'Financial returns, price determinants, and genre effects in American art investment', *Journal of Cultural Economics* 20(4), 359–383.
- Allison, J. Jr, Mark, T. B., Burdine, K. H. & Shockley, J. M. (2022), 'A hedonic analysis of factors impacting the value of planters on the used machinery market', *Agricultural and Resource Economics Review* 51(2), 266–282.
- Cross, R., Plantinga, A. J. & Stavins, R. N. (2011), 'The value of terroir: Hedonic estimation of vineyard sale prices', *Journal of Wine Economics* 6(1), 1–14.
- Cross, T. L. & Perry, G. M. (1995), 'Depreciation patterns for agricultural machinery', *American Journal of Agricultural Economics* 77(1), 194–204.
- Diekmann, F., Roe, B. E. & Batte, M. T. (2008), 'Tractors on eBay: Differences between internet and in-person auctions', *American Journal of Agricultural Economics* 90(2), 306–320.
- European Central Bank (2026), 'Euro foreign exchange reference rates', ECB Data Portal.
- Fettig, L. P. (1963), 'Adjusting farm tractor prices for quality changes, 1950–1962', *Journal of Farm Economics* 45(3), 599–611.
- Feuz, R. (2023), 'Hedonic price analysis of used tractors', *Applied Economics Teaching Resources* 5(1), 79–95.
- Garay, U., Puggioni, G., Molina, G. & ter Horst, E. (2022), 'A Bayesian dynamic hedonic regression model for art prices', *Journal of Business Research* 151, 310–323.
- Gelman, A. (2006), 'Prior distributions for variance parameters in hierarchical models', *Bayesian Analysis* 1(3), 515–533.
- Griliches, Z. (1961), 'Hedonic price indexes for automobiles: An econometric analysis of quality change', in *The Price Statistics of the Federal Government*, National Bureau of Economic Research, New York, pp. 173–196.
- Hadrich, J. C., Larsen, R. & Olson, F. E. (2013), 'Impact of the Section 179 tax deduction on machinery investment', *Agricultural Finance Review* 73(3), 458–468.
- Harrison, D. Jr & Rubinfeld, D. L. (1978), 'Hedonic housing prices and the demand for clean air', *Journal of Environmental Economics and Management* 5(1), 81–102.
- Kolbe, J., Schulz, R., Wersing, M. & Werwatz, A. (2021), 'Real estate listings and their usefulness for hedonic regressions', *Empirical Economics* 61(6), 3239–3269.
- Oczkowski, E. (1994), 'A hedonic price function for Australian premium table wine', *Australian Journal of Agricultural Economics* 38(1), 93–110.
- Pakes, A. (2003), 'A reconsideration of hedonic price indexes with an application to PC's', *American Economic Review* 93(5), 1578–1596.
- Pearson, M., Fiechter, C. M. & Campbell, A. (2026), 'A hedonic analysis of farm tractor auction prices', *Agricultural and Resource Economics Review* 55(2), 223–241.
- Perry, G. M., Bayaner, A. & Nixon, C. J. (1990), 'The effect of usage and size on tractor depreciation', *American Journal of Agricultural Economics* 72(2), 317–325.
- Perry, G. M. & Nixon, C. J. (1991), 'Optimal tractor replacement: What matters?', *Review of Agricultural Economics* 13(1), 119–128.

- Reid, D. W. & Bradford, G. L. (1983), ‘On optimal replacement of farm tractors’, *American Journal of Agricultural Economics* 65(2), 326–331.
- Rosen, S. (1974), ‘Hedonic prices and implicit markets: Product differentiation in pure competition’, *Journal of Political Economy* 82(1), 34–55.
- Rue, H., Martino, S. & Chopin, N. (2009), ‘Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations’, *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 71(2), 319–392.
- Rue, H., Riebler, A., Sørbye, S. H., Illian, J. B., Simpson, D. P. & Lindgren, F. K. (2017), ‘Bayesian computing with INLA: A review’, *Annual Review of Statistics and Its Application* 4, 395–421.
- Saito, W. & Nakatsuma, T. (2023), ‘Hierarchical Bayesian hedonic regression analysis of Japanese rice wine: Is the price right?’, *International Journal of Wine Business Research* 35(2), 256–277.
- Simpson, D. P., Rue, H., Martins, T. G., Riebler, A. & Sørbye, S. H. (2017), ‘Penalising model component complexity: A principled, practical approach to constructing priors’, *Statistical Science* 32(1), 1–28.
- Wilson, P. (2010), ‘Estimating tractor depreciation: The impact of choice of functional form’, *Journal of Farm Management* 13(12), 799–818.

Appendix

A Data pre-processing

The raw data were provided in the file `market_listings.csv`, which contained 354,607 records and 46 variables. As described in the main text, the original data covered several types of agricultural machinery and required substantial cleaning before they could be used for the analysis. Rather than importing all available fields, the data were read by selecting the 25 variables required for sample construction, price conversion and model estimation. These are reported in Table 8. Some of these fields were used only during data preparation and were not retained in the final statistical models. In particular, `currency` and `observed_date` were used to convert prices into euros, while `vat_basis` and `price_excl_vat` were used to construct VAT-exclusive asking prices. The listing identifier `market_id` was retained for auditing and data-ordering purposes.

Table 8: Variables retained from the raw dataset

Group	Variables
Identification and listing information	<code>market_id</code> , <code>channel</code> , <code>price_type</code> , <code>source</code> , <code>country</code> , <code>equipment_type</code> , <code>observed_date</code>
Price information	<code>price</code> , <code>currency</code> , <code>vat_basis</code> , <code>price_excl_vat</code>
Machine characteristics	<code>make</code> , <code>model_series</code> , <code>year</code> , <code>hours</code> , <code>horsepower</code>
Transmission and positioning equipment	<code>transmission</code> , <code>gps</code>
Binary equipment indicators	<code>front_pto</code> , <code>front_loader</code> , <code>front_linkage</code> , <code>front_axle_suspension</code> , <code>cab_suspension</code> , <code>air_conditioning</code> , <code>isobus</code>

A.1 Initial standardisation

Once the database was imported, empty strings and the values `NA`, `N/A` and `NULL` were treated as missing. Categorical entries were initially standardised by removing leading and trailing whitespace and replacing repeated internal spaces with a single space. The value `unknown` in the `vat_basis` variable was also recoded as missing.

A.2 Manufacturer names

The `make` variable contained a large number of inconsistencies. Differences in capitalisation, punctuation and formatting were first standardised. Clear spelling errors and abbreviations were then mapped to the corresponding manufacturer. Entries containing both the manufacturer and the model name were assigned to the manufacturer alone. The same procedure was applied to advertisement text or longer product descriptions when the manufacturer could still be identified unambiguously. Examples of the manual harmonisation are reported in Table 9. Values that did not identify a valid manufacturer were recoded as missing. These included generic equipment descriptions, unknown categories, promotional statements and entries referring only to a type of machinery or attachment. After this cleaning stage, valid manufacturers representing less than 0.03% of the full dataset were grouped into an *Other* category to reduce sparsity.

Table 9: Examples of manufacturer-name harmonisation

Type of inconsistency	Original entry	Cleaned value
Capitalisation	CLAAS	Claas
Capitalisation	DIECI	Dieci
Formatting	Deutz Fahr	Deutz-Fahr
Formatting	Case-IH	Case IH
Formatting	Mercedes Benz	Mercedes-Benz
Abbreviation	CAT	Caterpillar
Spelling error	Carerpillar	Caterpillar
Spelling error	John Derre	John Deere
Spelling error	Komatzu	Komatsu
Spelling error	Schaffer	Schäffer
Manufacturer and model combined	Ford 5000	Ford
Manufacturer and model combined	Fiat 880 DT/5	Fiat
Manufacturer and model combined	International 955 AXL	International
Product description	FENDT CARGO 4X75 COMPACT	Fendt
Advertisement text	INKOMMANDE TRAKTOR JOHN DEERE 1640	John Deere
Advertisement text	SÄNK! PRIS! New Holland	New Holland

A.3 Model-series identifiers

The `model_series` variable also showed considerable variation in formatting. Model descriptions were converted to lowercase and unnecessary whitespace was removed. A compact identifier was then created by deleting spaces, punctuation and other non-alphanumeric characters. This allowed formatting variants such as 6215 R, 6215-R and 6215R to be treated as the same model-series entry. Where the cleaned manufacturer name was also included in the model description, it was removed from the compact identifier. This avoided treating entries such as John Deere 6215R and 6215R as separate categories. Values that became empty after manufacturer removal were recoded as missing.

A.4 Transmission categories

The original `transmission` variable contained manufacturer-specific commercial names, multilingual descriptions, spelling variants, HTML elements and free text responses. Treating each original description as a separate category would have produced a large number of sparse or substantively overlapping levels. Transmission descriptions were therefore cleaned by removing HTML tags, converting accented characters to their Latin equivalents, converting text to lowercase and replacing punctuation with spaces. Regular-expression rules were then used to assign each entry to a broader technological family. Specific commercial names were classified before more general expressions. For example, systems such as *Vario*, *AutoPowr*, *TTV*, *Dyna-VT* and *Cmatic* were assigned to the CVT or stepless category. Labels including *AutoQuad*, *PowerQuad*, *Hexashift* and *Dyna-6* were assigned to the semi-powershift group. Descriptions that referred only to a powershift or automatic transmission, without enough information for a more precise classification, were retained in separate unspecified categories. Missing values were classified as *Not reported*, since the absence of transmission information in a listing does not identify the type of transmission installed. The categories used in the final dataset are reported in Table 10. Manual or mechanical transmission was used as the reference level in the statistical models. Some of these categories were no longer represented after the final sample-selection and complete-case cleaning steps. Table 10 therefore reports the full set of transmission categories defined during preprocessing, whereas the statistical models include only the levels present in the final analytical sample.

Table 10: Final transmission categories

Category	Description
Manual / mechanical	Manual, mechanical or synchronised transmission. This category was used as the reference level.
CVT / stepless	Continuously variable or stepless transmission, including manufacturer-specific commercial systems.
Full powershift	Transmission allowing powershift operation across the full range of gears.
Semi-powershift	Transmission allowing powershift operation within a limited number of ranges or groups of gears.
Powershift unspecified	A powershift transmission was reported, but the description did not allow it to be classified as either full or semi-powershift.
Hydrostatic	Hydrostatic transmission or an equivalent HST description.
Shuttle / reverser	Mechanical or hydraulic shuttle and reverser systems.
Automatic unspecified	An automatic transmission was reported without a more precise technological description.
Other / unclassified	The description could not be assigned reliably to one of the preceding technological categories.
Not reported	No transmission information was included in the listing.

A.5 GPS categories

The `gps` variable contained fewer distinct descriptions but still required some harmonisation. Entries were converted to lowercase and equivalent descriptions were grouped into common categories. The values `RTK` and `RTK GPS` were combined into the *RTK* category. The values `GPS receiver` and `receiver` were classified as *GPS receiver*, while `ready` and `GPS ready` were assigned to *GPS ready*. Generic affirmative responses, including `yes` and `ja`, were classified as *GPS present*, since they indicated that GPS equipment was reported but provided no information on the type of technology. Missing GPS values were converted to *Not reported*. This level was used as the reference category. Descriptions that could not be assigned to one of the final categories were placed in a temporary *Other / unclassified* group for inspection. This residual group was not retained as a category in the final modelling factor.

Table 11: Final GPS categories

Category	Description
Not reported	No GPS information was included in the listing. This category was used as the reference level.
GPS present	GPS equipment was reported, but its type was not specified.
GPS ready	The tractor was reported as prepared for the installation or use of GPS equipment.
GPS receiver	A GPS receiver was explicitly reported.
Auto steering	An automatic steering system was reported.
RTK	A real-time kinematic positioning system was reported.

A.6 Binary equipment variables

The seven binary equipment variables were recorded as `TRUE` when the corresponding feature was reported and as missing otherwise. Under the coding convention used in the original data, missing entries were interpreted as the feature not being reported as present and were recoded as `FALSE`. The variables treated in this way were `front_pto`, `front_loader`, `front_linkage`, `front_axle_suspension`, `cab_suspension`, `air_conditioning` and `isobus`.

A.7 Machine age and quantitative validity checks

Machine age was calculated as the difference between the calendar year in which the listing was observed and the reported year of manufacture:

$$\text{Age}_i = \text{year}(\text{observed_date}_i) - \text{year}_i.$$

Negative ages indicate that the reported year of manufacture was later than the year in which the listing was observed. These values were treated as inconsistent and recoded as missing. Broad validity rules were also applied to age, operating hours and horsepower. The purpose of these rules was to remove clearly invalid values without excluding observations merely because they were unusual. The thresholds are reported in Table 12.

A.8 Construction of the price variables

Two separate price variables were constructed to distinguish realised transaction prices from advertised asking prices. For observations classified as completed sales, `price_sold` was assigned the value recorded in the original `price` field. No adjustment for VAT was made to realised sale prices. These prices were subsequently converted from their original currencies into euros. For observations classified as asking-price listings, `price_ask` was constructed on a VAT-exclusive basis. When `vat_basis` was

Table 12: Validity rules applied to quantitative variables

Variable	Values recoded as missing	Reason
Age	Values below 0 or above 120 years	Inconsistent construction year information or an implausible machine age.
Operating hours	Values below 0 or greater than or equal to 100,000	Negative usage is impossible, while values of 100,000 hours or more were considered implausible for the machinery covered by the analysis.
Horsepower	Values less than or equal to 0 or above 2,500	Non-positive power is invalid, while values above 2,500 horsepower fall outside a plausible range for the equipment considered.

recorded as **net**, the original reported price was used. When **vat_basis** was recorded as **gross**, the corresponding value in **price_excl_vat** was used. Asking-price observations with missing or unknown VAT information could not be converted reliably to a common VAT basis and were therefore assigned a missing asking price. The original prices were reported in several currencies. To make observations comparable across countries, both realised sale prices and VAT-exclusive asking prices were converted into euros using daily reference exchange rates obtained from the European Central Bank Data Portal. The required currencies and date range were identified directly from the dataset. The download period began seven days before the earliest listing date, ensuring that a previous exchange rate was available if the first observations occurred during a weekend or public holiday. The ECB rates were expressed as units of foreign currency per euro. Observations already denominated in euros were assigned an exchange rate of one. ECB reference rates are not published on weekends and certain public holidays. A complete sequence of calendar dates was therefore created for each currency, and the most recently available previous exchange rate was carried forward to dates for which no new rate was published. The resulting daily exchange-rate dataset was saved separately to ensure that the currency conversion could be reproduced.

A.9 Construction of the analytical samples

Following currency conversion, separate sold-price and asking-price datasets were created. The sold-price dataset retained observations that were classified as completed sales, referred to tractors and had a finite, strictly positive sale price expressed in euros. The response variable in this dataset was named **price_sold**. The asking-price dataset retained observations classified as asking-price listings, referred to tractors and had a finite, strictly positive VAT-exclusive asking price expressed in euros. The response variable in this dataset was named **price_ask**. Both datasets retained the same cleaned machine characteristics and equipment variables. Fields used only during price construction, including the original prices, currency, VAT basis and exchange rates, were no longer required and were excluded. Complete-case versions of the two datasets were then produced by removing observations with at least one missing value among the retained variables. This operation did not remove original missing values in GPS or transmission when they had already been recoded as *Not reported*. Similarly, missing entries in the binary equipment fields had already been recoded as **FALSE**. Unused factor levels were dropped after sample selection. Consecutive integer identifiers were then constructed for manufacturer, model series and country for use in the group-level components of the INLA models.

Table 13: Final analytical samples

Dataset	Observations	First date	Last date	Distinct months
Sold-price dataset	4,235	2021-12-16	2026-06-11	48
Asking-price dataset	63,655	2019-07-11	2026-07-16	29

A.10 Chronological training and test split

The sold-price sample was divided chronologically into training and test sets. The target was to allocate approximately 80% of the observations to the training set and 20% to the test set. The data were first ordered by `observed_date` and `market_id`. The number of observations recorded on each date was then calculated. Among the possible cutoff dates, the selected cutoff was the one that produced a test-set size closest to 20% of the complete sold-price sample. All observations recorded before the cutoff date were assigned to the training set, while observations recorded on or after the cutoff date were assigned to the test set. Observations from the same date were therefore always assigned to the same sample. This chronological division was preferred to a random split because the objective was to evaluate the models on transactions observed later in time. The resulting test performance is consequently more representative of the intended forecasting setting. The cutoff date selected from the sold-price sample was also applied to the asking-price dataset. This ensured that sold and asking-price observations used for model estimation covered the same historical period. Since the temporal distributions of the two types of listings were not identical, the proportion of asking-price observations assigned to the test period was not necessarily exactly 20%.

Table 14: Chronological training and test samples

Dataset	Training	Test	Test share	Training period	Test period
Sold-price dataset	3,369	866	20.45%	2021-12-16 – 2026-03-28	2026-03-29 – 2026-06-11
Asking-price dataset	33,162	30,493	47.90%	2019-07-11 – 2026-03-28	2026-03-29 – 2026-07-16

Code and Data Access

The code will be made available at https://github.com/riccardomonforte7-lgtm/Dissertation_spectinga after personal information and copyrighted material have been removed and the file structure has been cleaned. The raw dataset will not be included in the repository, as it is subject to data-sharing agreements and cannot be made publicly available.

AI disclosure

Generative AI was used in a limited way to support the preparation of this report. In particular, ChatGPT was used to help with LaTeX formatting, grammar checking, wording improvements, and general guidance on the structure and organisation of the report.

Generative AI was not used to generate the empirical results, perform the statistical analyses, or produce the final interpretation of the findings. All code, figures, tables, model outputs and conclusions were checked, selected and edited by the author.

University of Edinburgh – Own Work Declaration

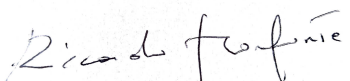
This sheet must be filled in, signed and dated - your work will not be marked unless this is done.

Name: Riccardo Monforte
Matriculation Number: S2882823
Title of work: Hierarchical Bayesian Hedonic Pricing of
Realised Wholesale Tractor Prices

I confirm that all this work is my own except where indicated, and that I have:

- Clearly referenced/listed all sources as appropriate
- Referenced and put in inverted commas all quoted text (from books, web, etc)
- Given the sources of all pictures, data etc. that are not my own
- Not made any use of the report(s) or essay(s) of any other student(s) either past or present
- Not sought or used the help of any external professional academic agencies for the work
- Acknowledged in appropriate places any help that I have received from others (e.g. fellow students, technicians, statisticians, external sources)
- Complied with any other plagiarism criteria specified in the Course handbook

I understand that any false claim for this work will be penalised in accordance with the University regulations (<https://teaching.maths.ed.ac.uk/main/msc-students/msc-programmes/statistics/data-science/assessment/academic-misconduct>).

Signatu 

Date 07/08/2026